Contents

1 The Baire Theorem and Applications

1.1 Baire Spaces

Theorem 1.1.1 (Baire)**.** *Let (E, d) be a complete metric space. $(O_n)_{n\geq 1}$ is a sequence of dense open sets in E. Then $O := \bigcap_{n\geq 1} O_n$ is dense in E.*

Remark. The requirement [open] of the O_ns can not be removed. For example, in $\mathbb{R}$, both $\mathbb{Q}$ and $\mathbb{Q}^c$ are dense, but their intersection is empty.

Definition 1.1.2. We say a topological space E is a **Baire space**, if the intersection of any countably many dense open sets in E is still dense in E.

Remark. This definiton implies that complete metric spaces are Baire space. However, this statement is not very rigorous, since [Barie space] is a pure topological concept, whereas the completeness of a metric space is not. Indeed, different metrics can induce the same topology, and one can be complete while the other may not. Therefore, a more accurate statement should be: If the topology of a topological space X can be induced by a complete metric, then X is a Baire space.

Remark. Note that the complement of a dense open set is a closed set with empty interior. Therefore, an equivalent definition of a Baire space, say E, is: the union of any countably many closed set with empty interior in E still has empty interior in E.

Theorem 1.1.3. *Let E be Baire space. Then*

1. *Any open subset of E is Baire.*
2. *Let $(F_n)_{n\geq 1}$ be a sequence of closed subset of E, and $E = \bigcup_{n\geq 1} F_n$. Then $\bigcup_{n\geq 1} \mathring{F}_n$ is dense in E.*

Proof. (1). Let Ω be an arbitrary open subset of E. To show that Ω is Baire, let $(O_n)_{n\geq 1}$ be a sequence of dense open sets in Ω. Since Ω is open, each O_n is also open in E. Let

$$U_n := O_n \cup \overline{\Omega}^c$$

U_n is open and dense in E, hence $(U_n)_{n\geq 1}$ is a sequence of open and dense subsets of E. Since E is Baire, $\bigcap_{n\geq 1} U_n$ is dense in E. But

$$\bigcap_{n\geq 1} U_n = \left(\bigcap_{n\geq 1} O_n\right) \cup \overline{\Omega}^c$$

This implies $\bigcap_{n\geq 1} O_n$ is dense in Ω, as desired.

(2). Pick any non-empty open subset Ω in E. Then

$$\Omega = \Omega \cap E = \bigcup_{n\geq 1} (\Omega \cap F_n)$$

Note that $\Omega \cap F_n$ is closed in Ω. From (1) we know that Ω is Baire, hence $\exists$ some n such that $(\Omega \cap F_n)^{\mathrm{o}} \neq \varnothing$ (interior taken in the subspace topology on Ω). But since Ω is open in E, the interior of $\Omega \cap F_n$ taken in Ω is the same as taken in E, hence $(\Omega \cap F_n)^{\mathrm{o}} \neq \varnothing$ (interior taken in E). That is, $\Omega \cap \mathring{F}_n \neq \varnothing$. Therefore, $\bigcup_{n\geq 1} \mathring{F}_n$ is dense in E. □

Definition 1.1.4. Let E be a topological space.

1. A set in E is called a $\mathcal{G}_\delta$ set if it is the intersection of countably many open subsets of E. A set in E is called an $\mathcal{F}_\sigma$ set if it is the union of countably many closed subsets of E.
2. A set $A \subseteq E$ is called **meagre** (or **of first category**) if A is contained in some $\mathcal{F}_\sigma$ set with empty interior. A set $A \subset E$ is called **residual** (or **comeagre**) if A contains a dense $\mathcal{G}_\delta$ subset of E.
3. A subset $A \subseteq E$ is **nowhere dense** if its closure has empty interior:

$$\mathring{\overline{A}} = \varnothing$$

   Equivalently, in every nonempty open set $U \subseteq E$, $\exists$ a nonempty open set $V \subseteq U$ such that $V \cap A = \varnothing$ (so A is "dense in no region").

Remark. Let E be a Baire space. Then the union of countably many meagre sets is still meagre. Hence, $A \subseteq E$ is meagre if and only if it is a countable union of nowhere dense sets.

Remark. We may think of meagre sets as an analogue of zero-measure sets. Their properties are very similar.

Theorem 1.1.5. *Let E be a Baire space, (F, δ) a metirc space. Let $(f_n)_n \subset C(E, F)$ be a sequence of mapping that converges to f pointwisely. Let*

$$\mathrm{Cont}(f) := \{x \in E : f \textit{ is continuous at } x\}$$

Then $\mathrm{Cont}(f)$ *is a dense* $\mathcal{G}_\delta$ *set in* E.

Proof. TBD ... $\square$

1.2 The Banach-Steinhaus Theorem

In this section and the following section, E and F are always normed vector spaces over a field $\mathbb{K}$.

Theorem 1.2.1 (Banach-Steinhaus). *Let E be a Banach space, F a normed vector space, $(u_i)_{i\in I}$ a collection of BLOP from E to F. If for every $x \in E$, $\sup_{i\in I}\|u_i(x)\| < \infty$, Then*

$$\sup_{i\in I}\|u_i\| < \infty$$

Proof. Let

$$M(x) := \sup_{i\in I}\|u_i(x)\| \qquad \text{and} \qquad F_n := \{x \in E : M(x) \leq n\}$$

Since u_i is continuous, $\{x \in E : \|u_i(x)\| \leq n\}$ is closed in E. Therefore,

$$F_n = \bigcap_{i\in I}\{x \in E : \|u_i(x)\| \leq n\}$$

is also closed in E. Now, by our assumption, for each $x \in E$, $M(x) < \infty$, hence

$$E = \bigcup_{n\geq 1} F_n$$

By Theorem 1.1.1, since E is Banach, E is Baire. Then, by Theorem 1.1.3, $\bigcup_{n\geq 1} \mathring{F}_n$ is dense in E. Hence, $\exists$ some $n \geq 1$ such that $\mathring{F}_n \neq \varnothing$. That means $\exists$ an open ball $B(x_0, r) \subset \mathring{F}_n$, which implies for any $x \in B(x_0, r)$

$$M(x) \leq n \quad \text{(i.e. } \|u_i(x)\| \leq n, \forall\, i \in I)$$

By translating to the origin, this is equivalent to: $\forall\, x \in B(0, r)$, $\|u_i(x_0 + x)\| \leq n$. Hence $\forall\, i \in I$,

$$\|u_i(x)\| \leq \|u_i(x_0 + x)\| + \|u_i(x_0)\| \leq n + M(x_0).$$

Denote $C := n + M(x_0)$. Then for all $i \in I$ and all $x \in B(0, r)$, we have $\|u_i(x)\| \leq C$. In particular, for any $y \in E$ with $\|y\| \leq 1$, we have $ry \in B(0, r)$, hence

$$\|u_i(y)\| = \frac{1}{r}\,\|u_i(ry)\| \leq \frac{C}{r}.$$

Taking the supremum over $\|y\| \leq 1$ gives $\|u_i\| \leq C/r$ for all $i \in I$. Therefore,

$$\sup_{i\in I}\|u_i\| \leq \frac{C}{r} < \infty.$$

□

Remark. For each point $x \in E, \sup_{i\in I}\|u_i(x)\| < \infty$, means that (u_i) is pointwisely bounded; Whereas $\sup_{i\in I}\|u_i\| < \infty$ means (u_i) is a collection of uniformly bounded linear operator. Therefore, we sometimes call Banach-Steinhaus uniform boundedness theorem.

Banach-Steinhaus was called " Principle of concentration of singularity" by the pioneers. It makes us understand this theorem better from the converse direction: If $(u_i)_{i\in I}$ is unbounded, then there exists a point $x \in E$ such that $(u_i(x))_{i\in I}$ is unbounded on F. In this sense, we may modify the theorem as follows:

Theorem 1.2.2. *Let E be a Banach space, F a normed vector space, $(u_i)_{i\in I}$ a collection of BLOP from E to F satisfying*

$$\sup_{i\in I}\|u_i\| = \infty$$

Then $G := \{x \in E : M(x) = \infty\}$ is a dense $\mathcal{G}_\delta$ set in E.

Proof. We have

$$G = \bigcap_{n\geq 1}\{x \in E : M(x) > n\} = \bigcap_{n\geq 1}\Omega_n$$

where $\Omega_n = \{x \in E : M(x) > n\}$. Let F_n be the set defined in the proof of Theorem 1.2.1, so $\Omega_n = F_n^c$, hence open. So G is a $\mathcal{G}_\delta$ set. To show G is dense, suppose that $\exists$ some n such that Ω_n is not dense in E. i.e., $\mathring{F}_n \neq \varnothing$. Then by the proof of Theorem 1.2.1, (u_n) is uniformly bounded, a contradiction. So Ω_n is dense in E for every n. Therefore, by Theorem 1.1.1, G is dense. $\square$

Corollary 1.2.3. *Let E be a Banach space, F a normed vector space, $(u_n)_{n\geq 1}$ a collection of BLOP from E to F. If $(u_n)_{n\geq 1}$ converges to u pointwisely, then $u \in \mathcal{B}(E, F)$ and*

$$\|u\| \leq \liminf_{n\to\infty}\|u_n\|$$

Proof. TBD ... $\square$

1.3 The Open Mapping Theorem and the Closed Graph Theorem

Theorem 1.3.1. *Let E and F be Banach spaces, $u \in \mathcal{B}(E, F)$. If $\mathrm{Im}(u)$ is not meagre in F, then*

1. *$\exists\, r > 0$ such that $rB_F \subset u(B_E)$ where B_E and B_F are unit open ball in E and F respectively. And u is surjective.*
2. *u is an open mapping.*

Proof. TBD ... $\square$

Corollary 1.3.2. *Let E and F be Banach spaces, $u \in \mathcal{B}(E, F)$ surjective. Then u is an open mapping and $\exists\, r > 0$ such that $rB_F \subset u(B_E)$.*

Proof. It suffices to show that $u(E)$ is not meagre in F. Note that $u(E) = F$ since u is surjective, and F is a Baire space since it is a Banach space, which is of course not a meagre set. $\square$

Corollary 1.3.3. *Let E and F be Banach spaces, $u : E \longrightarrow F$ is a continuous linear bijection. Then u^{-1} is also continuous.*

Proof. It suffices to show that u is an open mapping, which is clear by our previous corollary. $\square$

Theorem 1.3.4. *Let E and F be Banach spaces, $u : E \longrightarrow F$ is a linear map. Then u is continuous if and only if the graph $G(u)$ is closed.*

Proof. TBD ... $\square$

2 Hilbert Spaces

2.1 Inner Product Spaces

Definition 2.1.1. Let H be a vector space over $\mathbb{K}$ (where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$). A map

$$\langle \cdot, \cdot \rangle : H \times H \longrightarrow \mathbb{K}$$

is called an **inner product** on H if it satisfies the following properties:

1. **Bilinearity.**
   (a) If $\mathbb{K} = \mathbb{R}$, then $\langle \cdot, \cdot \rangle$ is **bilinear**.
   (b) If $\mathbb{K} = \mathbb{C}$, then $\langle \cdot, \cdot \rangle$ is **linear in the first variable** and **conjugate-linear in the second variable**. i.e.
   $$\langle x, \lambda y_1 + y_2 \rangle = \overline{\lambda} \langle x, y_1 \rangle + \langle x, y_2 \rangle, \qquad \forall\, y_1, y_2 \in H,\ \lambda \in \mathbb{C}.$$
2. **Conjugate symmetry.** For all $x, y \in H$,
   $$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$
3. **Positive definiteness.** For all $x \in H$, we have $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

In this case, we call $(H, \langle \cdot, \cdot \rangle)$ an **inner product space**.

Example 2.1.2. The following are some examples of inner product space:

1. $l_2^n = \mathbb{K}^n$. The inner product is defined to be: $\langle x, y \rangle = \sum_{k=1}^n x_k \overline{y_k}$, where $x = (x_1, ..., x_n), y = (y_1, ..., y_n)$.
2. l_2 space:
   $$l_2 = \left\{ x = (x_n)_{n \geq 1} \subset \mathbb{K} : \sum_{k=1}^{\infty} |x_k|^2 < \infty \right\}$$
   The inner product is defined to be: $\langle x, y \rangle = \sum_{k=1}^{\infty} x_k \overline{y_k}$
3. $L_2(\Omega, \Sigma, \mu)$. The inner product is defined to be:
   $$\langle f, g \rangle = \int_{\Omega} f(\omega) \overline{g(\omega)} \,\mathrm{d}\, \mu$$
4. $C([0,1]; \mathbb{K})$. The inner product is defined to be
   $$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} \,\mathrm{d}\, t$$

Remark. Let H be an innner product space. Define $\|x\| := \langle x, x \rangle^{1/2}$. Then, $\|\cdot\|$ is a norm on H. Indeed, by the properties of a inner product, here we only need to check the triangle inequality:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \overline{\langle x, y \rangle} + \langle y, y \rangle = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2 \end{aligned}$$

From this, we introduce the following concept:

Definition 2.1.3. Let H be an inner product space, let $\|x\| := \langle x, x\rangle^{1/2}$. We call $\|\cdot\|$ the **norm induced by the inner product**. Hence, H becomes a normed linear space. If the norm is complete, then H is called a **Hilbert space**.

Example 2.1.4. In example 2.1.2, l_2^n, l_2 and $L_2(\Omega, \Sigma, \mu)$ are all Hilbert spaces. However, $C([0,1]; \mathbb{K})$ is not. Its compeletion is $L_2(0,1)$.

2.2 Projection Operators

The following is the projection theorem for closed convex sets in Hilbert spaces.

Theorem 2.2.1. *Let H be a Hilbert space, $C \subseteq H$ a non-empty closed convex set. Then*

1. *$\forall x \in H, \exists!\, y \in C$ s.t.*
$$\|x - y\| = \operatorname{dist}(x, C)$$
This $y \in C$ is called the **projection** *of x on y, denoted as $P_C(x)$.*
2. *Let $y \in H$, then*
$$y = P_C(x) \iff \forall z \in C\,, \operatorname{Re}\langle x - y, z - y\rangle \le 0$$
3. *The map $x \mapsto P_C(x)$ is a Lipschitz map with $\lambda = 1$. That is, $\forall x, x' \in H$,*
$$\|P_C(x) - P_C(x')\| \le \|x - x'\|$$
4. *$P_C^2 = P_C$ and $P_C(H) = C$.*

Proof. (1). We first show exsistence. Fix an arbitrary $x \in H$, let $d = \operatorname{dist}(x, C)$. For any $k \in \mathbb{Z}_+$, we may pick $y_k \in C$ s.t.
$$d \le \|x - y_k\| \le d + 1/k$$
Now, it suffices to show that (y_k) is a Cauchy sequence. Because if so, then since H is complete and C is closed, (y_k) would converge to a point $y \in C$. Hence, by taking the limit, we get $\|x - y\| = d$ and so we are done. Note that:
$$\begin{aligned}\left(d + \frac{1}{n}\right)^2 + \left(d + \frac{1}{m}\right)^2 &\ge \|x - y_n\|^2 + \|x - y_m\|^2 \\ &= \frac{1}{2}\left(\|2x - (y_n + y_m)\|^2 + \|y_n - y_m\|^2\right) \\ &= \frac{1}{2}\left(4\left\|x - \frac{(y_n + y_m)}{2}\right\|^2 + \|y_n - y_m\|^2\right)\end{aligned}$$
Note that C is convex, so $\frac{(y_n+y_m)}{2} \in C$, hence $\left\|x - \frac{(y_n+y_m)}{2}\right\| \ge d$. Therefore,
$$\left(d + \frac{1}{n}\right)^2 + \left(d + \frac{1}{m}\right)^2 - 2d^2 \ge \frac{1}{2}\|y_n - y_m\|^2$$

Taking $n, m \to \infty$ yields $\|y_n - y_m\| \to 0$, as we disired. To show uniqueness, suppose $\exists$ another $y' \in C$ s.t. $\|x - y'\| = d$. Then

$$\begin{aligned}
2d^2 &= \|x-y\|^2 + \|x-y'\|^2 \\
&= \frac{1}{2}\left(\|2x-(y+y')\|^2 + \|y-y'\|^2\right) \\
&= \frac{1}{2}\left(4\left\|x - \frac{(y+y')}{2}\right\|^2 + \|y-y'\|^2\right) \\
&\geq 2d^2 + \frac{\|y-y'\|^2}{2}
\end{aligned}$$

Therefore, $\|y - y'\|^2 = 0$, and hence $y = y'$.

(2). $\Longrightarrow$: Let $y = P_C(x)$. Fix an arbitrary point $z \in C$, let $z' = \lambda y + (1-\lambda)z$. Since C is convex and $y, z \in C$, $z' \in C$. Therefore

$$\begin{aligned}
\|x-y\|^2 &\leq \|x-z'\|^2 \\
&= \|x-y-(z'-y)\|^2 \\
&= \|x-y\|^2 + \|z'-y\|^2 - 2\operatorname{Re}\langle x-y, z'-y\rangle
\end{aligned}$$

This implies $2\operatorname{Re}\langle x-y, z-y\rangle \leq (1-\lambda)\|z-y\|^2$. Taking $\lambda \to 1$ yields $\operatorname{Re}\langle x-y, z-y\rangle \leq 0$. $\Longleftarrow$: Fix an arbitrary $z \in C$, since $\operatorname{Re}\langle x-y, z-y\rangle \leq 0$, we have

$$\begin{aligned}
\|x-z\|^2 &= \|x-y-(z-y)\|^2 \\
&= \|x-y\|^2 + \|z-y\|^2 - 2\operatorname{Re}\langle x-y, z-y\rangle \\
&\geq \|x-y\|^2
\end{aligned}$$

Since $z \in C$ is taking arbitrarily, we have $\|x - y\| = \operatorname{dist}(x, C)$, as desired.

(3). Let $y = P_C(x), y' = P_C(x')$. Then

$$\begin{aligned}
\|y-y'\|^2 &= \langle y-y', y-y'\rangle \\
&= \langle y-x, y-y'\rangle + \langle x-x', y-y'\rangle + \langle x'-y', y-y'\rangle
\end{aligned}$$

Taking real parts on both sides yields:

$$\begin{aligned}
\|y-y'\|^2 &= \operatorname{Re}\langle y-x, y-y'\rangle + \operatorname{Re}\langle x-x', y-y'\rangle + \operatorname{Re}\langle x'-y', y-y'\rangle \\
&\leq \operatorname{Re}\langle x-x', y-y'\rangle \\
&\leq |\langle x-x', y-y'\rangle| \leq \|x-x'\|\,\|y-y'\|
\end{aligned}$$

Therefore, $\|y - y'\|^2 \leq \|x - x'\|$, as we desired.

(4) can be direcrly read off from the definition. $\square$

Definition 2.2.2. Let H be an inner product space. If $\langle x, y\rangle = 0$ for $x, y \in H$, then x and y are said to be orthogonal, denoted by $x \perp y$. For any subset $A \subset H$, define

$$A^{\perp} = \{x \in H : x \perp y,\ \forall y \in A\}.$$

The set $A^{\perp}$ is called the **orthogonal complement** of A in H.

Remark. Let H be an inner product space. Then

1. $x \perp y \implies \|x+y\|^2 = \|x\|^2 + \|y\|^2$
2. If $x, y \neq 0$, we can define
$$\cos\theta = \frac{\operatorname{Re}\langle x, y\rangle}{\|x\|\,\|y\|}, \qquad \theta \in [0, \pi]$$
3. Since the inner product is continuous, for any $A \subset H$, $A^\perp$ is a closed subspace and
$$A^\perp = \overline{A}^\perp = (\operatorname{span} A)^\perp = (\overline{\operatorname{span} A})^\perp$$

Note that a subspace of a Hilbert space H is not necessarily closed. Here is an example:

Example 2.2.3. Let $H = l_2$ over $\mathbb{R}$. Let
$$E := \{x = (x_n)_{n\geq 1} \in H : x_i = 0 \text{ for all but finitely many } i\}$$
Now $E \subsetneq H$ is a linear subspace. Consider a sequence $\{x^{(k)}\}_{k\geq 1}$ in E where
$$x_i^{(k)} = \begin{cases} \frac{1}{2^i} & \text{if } i \leq k \\ 0 & \text{otherwise} \end{cases}$$
Then this is a Cauchy sequence that converges in H, but its limit does not belong to E. In fact, in this case, $\overline{E} = H$.

When the closed convex set C in Theorem 2.2.1 is a subspace of H, the projection $P_C(-)$ has the following simple characterization:

Theorem 2.2.4. *Let H be a hilbert space, E be a closed subspace. Then for any $x \in H$, the projection $P_E(x) \in E$ is the only element in E satisfying $x - P_E(x) \perp E$. Moreover, P_E is a linear operator $\in \mathcal{L}(E, F)$, and $\|P_E\| \leq 1$.*

Proof. By Theorem 2.2.1 (2) we know:
$$\begin{aligned} y = P_E(x) &\iff \operatorname{Re}\langle x - y, z - y\rangle \leq 0, \forall z \in E \\ &\iff \operatorname{Re}\langle x - y, (z' + y) - y\rangle \leq 0, \forall z' \in E \\ &\iff \operatorname{Re}\langle x - y, \lambda z''\rangle \leq 0, \forall z'' \in E, \forall \lambda \in \mathbb{K} \\ &\iff \operatorname{Re}\overline{\lambda}\langle x - y, z\rangle \leq 0, \forall z \in E, \forall \lambda \in \mathbb{K} \end{aligned}$$
Convince yourself that the above $\iff x - y \perp E$. Hint: consider $\mathbb{K} = \mathbb{R}$ and $\mathbb{K} = \mathbb{C}$ separately. Then, we show that P_E is linear. Take any $x, x' \in H$, we have
$$(x + x') - (P_E(x) + P_E(x')) = (x - P_E(x)) + (x' - P_E(x')) \perp E$$
By the uniqueness of the projection, $P_E(x) + P_E(x') = P_E(x + x')$. For any $\lambda \in \mathbb{K}, x \in H$ (we may assume $\lambda \neq 0$), we have
$$x - P_E(x) \perp E \iff \lambda x - \lambda P_E(x) \perp E$$
By the uniqueness of the projection, $\lambda P_E(x) = P_E(\lambda x)$. Finally, $\|P_E\| \leq 1$ can be directly read off from Theorem 2.2.1 (3). $\square$

The above theorem gives the following two important corollaries, which are key tools for constructing normed orthogonal basis.

Corollary 2.2.5 (Orthogonal Decomposition)**.** *Let H be a Hilbert space. E is a subspace of H. Then*

$$H = \overline{E} \oplus E^{\perp}$$

Proof. Take any $x \in H$. Let $y = P_{\overline{E}}(x) \in \overline{E}$. Then $z := x - y \perp \overline{E}$, hence $z \in \overline{E}^{\perp} = E^{\perp}$. That is, $\exists$ a decomposition $x = y + z$ where $y \in \overline{E}$ and $z \in E^{\perp}$. Uniqueness: suppose $\exists\,(y', z') \in \overline{E} \times E^{\perp}$ s.t. $x = y' + z'$, then $y - y' = z' - z =: \theta$ and $\theta \in \overline{E} \cap E^{\perp}$. Hence $\theta = 0$. $\square$

Corollary 2.2.6. *Let H be a Hilbert space. E is a subspace of H. Then*

$$(E^{\perp})^{\perp} = \overline{E}.$$

Proof. Note that $E \subset (E^{\perp})^{\perp}$, takeing the closure on both sides yields $\overline{E} \subset \overline{(E^{\perp})^{\perp}} = (E^{\perp})^{\perp}$. Conversely, take any $x \in (E^{\perp})^{\perp}$, then $P_{E^{\perp}}(x) = 0$. By Corollary 2.2.5 we have:

$$x = P_{\overline{E}}(x) + P_{E^{\perp}}(x) = P_{\overline{E}}(x)$$

Hence $x \in \overline{E}$, as desired. $\square$

Remark. Let H be a Hilbert space. A is a subset of H. Then

$$(A^{\perp})^{\perp} = \overline{\operatorname{span} A}$$

This is because $A^{\perp} = (\overline{\operatorname{span} A})^{\perp}$.

2.3 Duality and Adjoint

It is hard to characterize the dual space of a normed linear space in general. However, the following Riesz Representation Theorem says that, for Hilbert spaces, the dual spaces are themselves.

Theorem 2.3.1 (Riesz Representation Theorem)**.** *Let H be a Hilbert space over $\mathbb{K}$, let $\phi : H \longrightarrow \mathbb{K}$. Then*

$$\phi \in H^* = \mathcal{B}(H, \mathbb{K}) \iff \exists\, y \in H \text{ s.t. } \forall\, x \in H,\ \phi(x) = \langle x, y \rangle$$

Moreover, such y is unique and $\|y\| = \|\phi\|$.

Proof. $\Longleftarrow$: Let $y \in H$. Define $\phi_y : H \longrightarrow \mathbb{K}$ by $\phi_y(x) = \langle x, y \rangle$. Clearly ϕ_y is linear. It remains to show that ϕ_y is bounded. By Cauchy-Schwarz:

$$|\phi_y(x)| \leq \|x\|\,\|y\|$$

Hence $\phi \in \mathcal{B}(H, \mathbb{K})$. If $y = 0$, then $\|\phi_y\| = 0$. Otherwise, let $x = \frac{y}{\|y\|}$. Then

$$|\phi_y(x)| = \left| \left\langle \frac{y}{\|y\|}, y \right\rangle \right| = \|y\|$$

which still gives $\|\phi_y\| = 0$.

$\Longrightarrow$: Let $\phi \in H^*$ and $E := \ker \phi$, so $E = \overline{E}$ since ϕ is continuous. Consequently, $H = E \oplus E^\perp$. If $\phi = 0$, then $y = 0$ will satisfy the condition. Hence, we may assume that $\phi \neq 0$, then $E \neq H$. So $\exists\, e \in E^\perp$ and $\phi(e) \neq 0$. By linearity we may assume $\phi(e) = 1$. Now, for any $x \in H$, we may write x as:

$$x = (x - \phi(x)e) + \phi(x)e =: x' + \phi(x)e$$

Note that $\phi(x') = 0$, hence $x' \in E$. Therefore

$$\langle x, e \rangle = \langle x' + \phi(x)e, e \rangle = \phi(x) \|e\|^2$$

Hence

$$\phi(x) = \frac{\langle x, e \rangle}{\|e\|^2} = \left\langle x, \frac{e}{\|e\|^2} \right\rangle$$

as desired. Now, it remain to show the uniqueness: Suppose $\exists\, y' \in H$ s.t.

$$\phi(x) = \langle x, y' \rangle, \forall x \in H$$

Then we have

$$0 = \langle x, y' \rangle - \langle x, y \rangle = \langle x, y' - y \rangle \ \forall x \in H$$

But this implies $y - y' = 0$, as desired. $\square$

Remark. The Riesz Representation Theorem shows that we may regard H^* and H as the same space in the isometric sense; that is,

$$y \in H \iff \phi_y \in H^*$$

And $\|y\| = \|\phi\|_y$. Moreover, we may define:

$$\begin{aligned} \Phi : \ H &\longrightarrow H^* \\ y &\longmapsto \phi_y \end{aligned}$$

that satisfies:

1. $\Phi(y + y') = \phi_u + \phi_{y'}$
2. For any $\lambda \in \mathbb{K}$ ($\mathbb{R}$ or $\mathbb{C}$), we have: $\Phi(\lambda y) = \overline{\lambda}\phi_y$

This implies Φ is linear if $\mathbb{K} = \mathbb{R}$, conjugate linear if $\mathbb{K} = \mathbb{C}$.

Theorem 2.3.2. *Let H and K be Hilbert spaces, $u \in \mathcal{B}(H, K)$. Then $\exists!\, u^* \in \mathcal{B}(K, H)$ s.t. for any $x \in K, y \in H$*

$$\langle u^*(x), y \rangle = \langle x, u(y) \rangle \tag{1}$$

Moreover, $\|u^*\| = \|u\|$. We call u^* the **adjoint** *of u.*

Proof. Fix any $x \in K$. Since $u \in \mathcal{B}(H, K)$ and the continuity of the inner product, the map:

$$\phi : H \longrightarrow \mathbb{K}$$

$$y \longmapsto \langle u(y), x\rangle$$

is continuous and linear. That is, $\phi \in H^*$. Then by Riesz, $\exists!\, z \in H$ s.t. for any $y \in H$:

$$\phi(y) = \langle u(y), x\rangle = \langle y, z\rangle$$

Now, setting $u^*(x) = z$ yields

$$\langle y, u^*(x)\rangle = \langle u(y), x\rangle$$

By symmetry, (1) holds. Convince yourself that u^* is linear. Now, by Riesz,

$$\begin{aligned}
\|u^*\| &= \sup_{x\in K, \|x\|\le 1} \|u^*(x)\| \\
&= \sup_{x\in K, \|x\|\le 1} \|\phi\| \\
&= \sup_{x\in K, \|x\|\le 1} \sup_{y\in H, \|y\|\le 1} |\langle u(y), x\rangle| \\
&= \sup_{y\in H, \|y\|\le 1} \|u(y)\| \\
&= \|u\|
\end{aligned}$$

as desired. The uniqueness of u^* is clear. □

Remark. We have the following results:

1. Let $u \in \mathcal{B}(H, K)$. Then $(u^*)^* = u$.
2. The map

$$\theta : \mathcal{B}(H, K) \longrightarrow \mathcal{B}(K, H)$$

$$u \longmapsto u^*$$

   is an isometry (i.e. isometric bijection). Moreover, θ is linear / conjugate linear if $\mathbb{K} = \mathbb{R}/\mathbb{C}$.
3. Let H, K, J be Hilbert spaces. $u \in \mathcal{B}(H, K), v \in \mathcal{B}(K, J)$. Then

$$(vu)^* = u^*v^* \in \mathcal{B}(J, H)$$

   Consequently, if u is invertible, then u^* is also invertible (let $v = u^{-1}$) and

$$(u^*)^{-1} = (u^{-1})^*$$

Definition 2.3.3. Let $u \in \mathcal{B}(H)$. If $u \circ u^* = u^* \circ u = \mathrm{id}_H$, then we call u a **unitary operator**.

Remark. All the unitary operators in $\mathcal{B}(H)$, with compositions, form a group. For example, when H is a finite-dimensional Hilbert space, i.e. $\dim H = n < \infty$, let $(e_1, \ldots, e_n)$ be an orthonormal basis of H (see the definition of orthonormal bases in the next subsection). For any $u \in \mathcal{B}(H)$, let $[u]$ denote the matrix of u with respect to the basis $(e_1, \ldots, e_n)$. Then

$$[u^*] = [u]^*$$

where $[u]^*$ is the conjugate transpose of $[u]$. The operator u is unitary if and only if $[u]$ is a unitary matrix.

2.4 Orthogonal Basis

We shall prove that every Hilbert space admits an orthonormal basis; this is a distinctive property of Hilbert spaces. Orthonormal bases are an effective tool for working with Hilbert spaces. Throughout this section, H will denote a Hilbert space.

Definition 2.4.1. Let I be an index set and let $(e_i)_{i\in I}$ be a family of vectors in H.

1. If for any $i, j \in I$ with $i \neq j$, we have $e_i \perp e_j$, then the family $(e_i)_{i\in I}$ is called **orthogonal**.
2. If $(e_i)_{i\in I}$ is orthogonal and, for every $i \in I$, $\|e_i\| = 1$, then $(e_i)_{i\in I}$ is called **orthonormal**.
3. If $\mathrm{span}((e_i)_{i\in I})$ is dense in H, then the family $(e_i)_{i\in I}$ is called **total**.
4. If $(e_i)_{i\in I}$ is orthonormal and total, then $(e_i)_{i\in I}$ is called an **orthonormal basis** of H.

Remark. Note that orthonormal basis is not unique.

Following is a diagram showing the relation between the above concepts. $A \longrightarrow B$ means that B requires A, as a prerequisite. (not logically implied by).

$$\begin{tikzcd}
 & \text{Orthonormal Basis} & \\
\text{Orthonormal} \arrow[ur] & & \text{Total} \arrow[ul] \\
 & \text{Orthogonal} \arrow[ul] &
\end{tikzcd}$$

Theorem 2.4.2. *Let* $(\mathbf{e}_1, ..., \mathbf{e}_n)$ *be an orthonormal set in* H*.* $E := \mathrm{span}(\mathbf{e}_1, ..., \mathbf{e}_n)$*. Then* $\forall\, \mathbf{x} \in H$*:*

$$P_E(\mathbf{x}) = \sum_{i=1}^{n} \langle \mathbf{x}, \mathbf{e}_i \rangle \, \mathbf{e}_i \qquad \textit{and} \qquad \|\mathbf{x}\|^2 = \|\mathbf{x} - P_E(\mathbf{x})\|^2 + \sum_{i=1}^{n} |\langle \mathbf{x}, \mathbf{e}_i \rangle|^2$$

Proof. Let $\mathbf{y} := \sum_{i=1}^{n} \langle \mathbf{x}, \mathbf{e}_i \rangle \, \mathbf{e}_i \in E$. Note that for each $k \in [n]$

$$\langle \mathbf{x} - \mathbf{y}, \mathbf{e}_k \rangle = 0$$

Hence $\mathbf{x} - \mathbf{y} \perp E$. So by Theorem 2.2.4, $\mathbf{y} = P_E(\mathbf{x})$. And since $(\mathbf{e}_i)$ is orthonormal, $\mathbf{y}^2 = \sum_{i=1}^{n} |\langle \mathbf{x}, \mathbf{e}_i \rangle|^2$ and

$$\|\mathbf{x}\|^2 = \|\mathbf{x} - \mathbf{y}\|^2 + \|\mathbf{y}\|^2 = \|\mathbf{x} - P_E(\mathbf{x})\|^2 + \sum_{i=1}^{n} |\langle \mathbf{x}, \mathbf{e}_i \rangle|^2$$

$\square$

Theorem 2.4.3. *Let $(\mathbf{e}_n)_{n\geq 1}$ be an orthonormal sequence in H. The followings are equivalent:*

1. $(\mathbf{e}_n)_{n\geq 1}$ *is total (and hence an orthonormal basis in H).*
2. $\forall\, \mathbf{x} \in H$, $\exists!$ *sequence (x_n) in $\mathbb{K}$ s.t.*

$$\sum_{n\geq 1} x_n \mathbf{e}_n \longrightarrow \mathbf{x}$$

Proof. $2 \implies 1$: This is clear since $2 \implies E := \operatorname{span}((\mathbf{e}_n)_{n\geq 1})$ is dense in H.

$1 \implies 2$: TBD... □

Theorem 2.4.4 (Bessel). *Let $(\mathbf{e}_n)_{n\geq 1}$ be an orthonormal sequence in H. Then for any $\mathbf{x} \in H$:*

$$\sum_{n\geq 1} |\langle \mathbf{x}, \mathbf{e}_n \rangle|^2 \leq \|x\|^2$$

Theorem 2.4.5 (Parseval). *Let $(\mathbf{e}_n)_{n\geq 1}$ be an orthonormal basis in H. Then for any $\mathbf{x} \in H$:*

$$\sum_{n\geq 1} |\langle \mathbf{x}, \mathbf{e}_n \rangle|^2 = \|x\|^2$$

More generally, for any $\mathbf{x}, \mathbf{y} \in H$:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n\geq 1} \langle \mathbf{x}, \mathbf{e}_n \rangle \langle \mathbf{e}_n, \mathbf{y} \rangle$$

Theorem 2.4.6. *Let H be a non-zero separable hilbert space, then H has a orthonormal basis*

Proof. Let $(\mathbf{a}_n)_{n\geq 1}$ be a countable dense subset of H. Starting from $\mathbf{a}_1$, remove all the vectors that can be spanned by the previous ones. This yields a new series of vectors $(\mathbf{v}_n)_{n\geq 1}$, which □